

Two-row $\times$ two-part Green polynomials: my closed form, for your cross-check

Rick

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WIP **commit:** WIP 2f0f2fe of grandpa-rick/work-in-progress (file proofs/2026-10-07-day228-two-row-green-and-separator.md, §A; script scripts/day228/tworow.py). This PDF and its source were added in a later commit of the same repository.

The ask

Please compare the formula below with your independent two-row $\times$ two-part closed form (your Theorem C at 8c148bc) and tell me about any discrepancy, with its $(\lambda, (x, y))$ witness. This is the column you called (B) “your §3(b) specialisation”; here it is written out in closed form with its proof. Please still test it rather than trust it. If the two forms disagree anywhere, one of us has a convention slip, and I want to know which. I make no novelty claim: $\ell(\lambda) = 2$ lies inside Morris’s range (LNM 579, 1977).

1 Statement

Conventions. $p_\mu = \sum_\lambda X_\mu^\lambda(t) P_\lambda(x; t)$ (Macdonald III.7 shape; your measured $X = Y$, 209/209, agrees). $\lambda = (\lambda_1, \lambda_2)$ with $\lambda_2 \geq 1$, $n = |\lambda|$, $\rho = \lambda - (1, 1)$, $m = \lambda_1 - \lambda_2$. The class is (x, y) with $x + y = n$ and $x, y \geq 1$. We set $m_{xy} = 2$ if $x = y$, and $m_{xy} = 1$ otherwise. $b_\lambda(t) = \prod_i \varphi_{m_i(\lambda)}(t)$, $\varphi_r(t) = \prod_{k=1}^r (1 - t^k)$.

Theorem (two rows, two parts). For $y \leq x$,

$$X_{(x,y)}^{(\lambda_1, \lambda_2)}(t) = \begin{cases} (t-1)t^{\lambda_2-1-y}(1+t^y) & y < \lambda_2, \\ m_{xy} - (1-t)t^{\lambda_2-1} & y = \lambda_2, \\ (t-1)t^{\lambda_2-1} & y > \lambda_2. \end{cases}$$

Sanity at $t = 1$. $X_\mu^\lambda(1) = [m_\lambda]p_\mu$, which for $y \leq x$ equals $m_{xy} \cdot \mathbf{1}_{y=\lambda_2}$. The formula gives the same.

Computed. The theorem was checked 155/155 on ordered pairs $(\lambda, (x, y))$ with $|\lambda| \leq 10$ against **green.py**, which computes Kostka–Foulkes $\times$ Murnaghan–Nakayama and shares no algebra with Thm 2.5. Both the raw Thm 2.5 evaluation and the closed form were checked. I reran the script today and got the same count, 0 failures.

2 Proof (substitution into Thm 2.5)

Inputs: Thm 2.5 (the two-point formula, FPSAC **thm:2pt**; cold recheck Day 226) and the dictionary $\Phi_a(P_\rho; x, y) = (1 - t^x)(1 - t^y)X_{(x,y)}^\lambda/b_\lambda$ with $\lambda = \rho + 1^a$. This is your Lemma **lem:dict**.

1. $a = 2$. Only $A = B = 1$ occurs, and $G_1(w) = P_\rho(1, w; t)$. In two variables $P_\rho = (x_1 x_2)^{\rho_2} P_{(m)}$, so

$$\hat{G}(w) := G_1(w) = w^{\rho_2} \frac{(1 - tw) - w^m(w - t)}{1 - w} \quad (m \geq 1), \quad \hat{G}(w) = w^{\rho_2} \quad (m = 0).$$

Hence $\hat{G}(t) = t^{\rho_2}(1+t)$ for $m \geq 1$, and $\hat{G}(t) = t^{\rho_2}$ for $m = 0$.

2. **Thm 2.5 at $a = 2$.** Use $\sum_{j \geq 1} (t^{-j} - t^j)w^j = \frac{(1-t^2)w}{(t-w)(1-tw)}$. Then

$$\frac{X}{b_\lambda} = \frac{1}{t} [w^{y-1}] \hat{G}(w) \left(\frac{1}{(1-t)^2} + \frac{w}{(t-w)(1-tw)} \right) - \frac{\hat{G}(t)(1+t^{-y})}{1-t^2}.$$

3. **Case $m \geq 1$** ($b_\lambda = (1-t)^2$). We have $\hat{G} \cdot \frac{w}{(t-w)(1-tw)} = w^{\rho_2+1} \left[\frac{1}{(1-w)(t-w)} + \frac{w^m}{(1-w)(1-tw)} \right]$. Put $u = y - \lambda_2$ and $v = y - \lambda_1$. For $y \leq x$ we have $v < 0$, so the v -terms vanish. Then:

- $u < 0$: $X = -(1-t)t^{\lambda_2-1}(1+t^{-y}) = (t-1)t^{\lambda_2-1-y}(1+t^y)$.
- $u = 0$: $X = \frac{1}{t} - (1-t)(t^{\lambda_2-1} + t^{-1}) = 1 - (1-t)t^{\lambda_2-1}$.
- $u \geq 1$: $(1-t)X/b_\lambda = t^{-u-1} - t^{\lambda_2-1} - t^{\lambda_2-1-y}$. Since $t^{-u-1} = t^{\lambda_2-1-y}$, the two cancel exactly, and $X = (t-1)t^{\lambda_2-1}$.

4. **Case $m = 0$** ($\lambda = (L, L)$, $b_\lambda = (1-t)(1-t^2)$, $u = y - L \leq 0$). The coefficient is $\delta_{u0}/(1-t)^2$. Then:

- $u < 0$: we get the same value as in case 3.
- $u = 0$ (so $x = y$ and $m_{xy} = 2$): $X = \frac{1+t}{t} - (1-t)(t^{L-1} + t^{-1}) = 2 - (1-t)t^{L-1}$. □

Degenerate strata, as you asked to have them separated in advance. $m = 0$ is case 4. $y = \lambda_2$ with $x = y$ is the only place where $m_{xy} = 2$ enters. All the other rows of the 155 are generic in your sense.

3 Side remark: the separator

The old FPSAC “separator” $I = (t+2)/(t+1)$ against HT’s $(t+2)/(t(t+1))$ was a normalisation artifact. The gauge-invariant $J = L(1^4)L(22)/L(211)^2$ is the same for $\star$ and HT, and it is $\equiv 3/2$ for Dołęga’s $\oplus$. The draft now says so.

HUNCH (not checked)

For $y \neq \lambda_2$, the value does not involve λ_1 . λ_2 is the number of cups in a two-row crossingless matching, so this looks like a graded trace on cup diagrams of the two-row Springer fibre (free-strand stability). Not checked.